

Shivam Raj

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SKILLS

Languages:	C, C++, Java, Python, JavaScript, TypeScript
Backend:	Spring Boot, Node.js, Express, FastAPI, REST APIs, JWT Authentication
Frontend:	HTML, CSS, React.js
Database:	MongoDB, PostgreSQL
Machine Learning:	Scikit-learn, MediaPipe, Computer Vision, Feature Engineering
Tools:	Git, Docker, Linux, AWS EC2, Swagger/OpenAPI
Frameworks:	Node, Spring
Soft Skills:	Articulation, Organizer, Critical Thinking, Leadership

PROJECTS

WriteFlow | [GitHub](#) | [Project Link](#)

Apr'2026 - Present

- Built a **full-stack blog platform backend** using **Spring Boot** with a **decoupled REST API architecture** supporting **posts, comments, and user management**.
- Designed **secure backend services** with **JWT-based authentication, role-based access control (RBAC), rate limiting, and input validation** following **OWASP best practices**.
- Implemented features including **pagination, profile management, and content workflows** using **Spring Data JPA** with **optimized database handling (PostgreSQL/H2)**.
- Deployed the application using **Docker-based containerization** with **Nginx reverse proxy**, enabling **seamless frontend-backend integration** and **scalable system design**.
- Tech Stack:** React, Spring Boot, PostgreSQL, Java, JWT, Docker, Nginx

RepMate | [GitHub](#) | [Project Link](#)

Jun'2025 - Present

- Built a **real-time fitness tracking system** using **computer vision** (MediaPipe) to detect body keypoints and **Machine learning** to classify exercise form with **probabilistic scoring**.
- Designed a **feature engineering pipeline** from **pose keypoints** (joint angles, velocity, temporal smoothing) for **sequence-based** rep detection and form evaluation.
- Integrated a **FastAPI-based ML inference module** with **WebSocket communication** into a **Spring Boot backend** with **JWT-based authentication** for low-latency real-time feedback.
- Implemented analytics features including rep tracking, performance metrics, and gamification, achieving **~96% accuracy** in controlled test scenarios.
- Tech Stack:** React, Spring Boot, PostgreSQL, Python, FastAPI, MediaPipe, OpenCV, Scikit Learn

Gym Management | [GitHub](#) | [Project Link](#)

Feb' 2026

- Built a **full-stack gym management platform** using **Spring Boot** and React with a decoupled **REST API architecture** for handling member lifecycle, subscriptions, and administrative workflows.
- Designed secure backend services with **JWT-based authentication, role-based access control (RBAC), rate limiting, and input validation** to ensure system security.
- Implemented core modules including membership management, trainer allocation, attendance tracking, and financial records with **PostgreSQL** and optimized query handling.
- Deployed the application using **containerized** backend services and cloud hosting, with API documentation via **Swagger** and real-time dashboard analytics supported by **caching mechanisms**.
- Tech Stack:** React, Spring Boot, PostgreSQL, Java, JWT

FinTrackAI | [GitHub](#) | [Project Link](#)

Jul' 2025

- Built a full-stack financial analytics platform using the **MERN stack** with secure **JWT-based authentication** and **Google OAuth** for user management.
- Developed automated expense tracking and data processing pipelines supporting **CSV/PDF ingestion, validation, and structured storage** using **MongoDB**.
- Integrated **AI-driven financial insights** using external models and designed a **classification pipeline** using text features (**TF-IDF**) with scope for ML-based expense categorization.
- Implemented **interactive dashboards** for real-time visualization of spending patterns, savings trends, and categorized financial insights.
- Tech Stack:** MongoDB, Express.js, React.js, Node.js, Scikit Learn, REST APIs

TRAINING

FLAMES '25 MERN with Gen AI W3grads Certificate	Jul'2025
- Resolved development bottlenecks in AI workflows and database performance by optimizing pipelines and debugging inconsistencies, improving system reliability and deployment stability. - Improved team efficiency by establishing a structured Git workflow and enforcing daily SCRUMs, reducing merge conflicts and ensuring consistent collaboration across distributed contributors. - Enhanced application scalability by addressing deployment and integration challenges in full-stack systems, enabling smoother releases and faster iteration cycles. - Built and refined a full-stack MERN application (FinTrackAI) during training, contributing to frontend UI/UX development, API integration, and end-to-end feature implementation. - Integrated Gemini API-based AI features and validated backend services using Postman, ensuring robust API performance, accurate data flow, and reliable system behavior. - Engineered AI-enabled features using Gemini API and optimized API testing workflows with Postman, improving debugging efficiency and ensuring high reliability of backend services.	

CERTIFICATES

Master Generative AI & Generative AI tools (Infosys) Certificate	Aug' 2025
Build Generative AI Apps and Solutions with No-Code Tools Certificate	Aug' 2025
Java Programming (iamneo) Certificate	May' 2025
Data Structures and Algorithms with C++ (iamneo) Certificate	Dec' 2024
The Bits and Bytes of Computer Networking (Google) Certificate	Sep' 2024
Computer Communications (University of Colorado) Certificate	Sep' 2024
TCP/IP and Advanced Topics (University of Colorado) Certificate	Sep' 2024

ACHIEVEMENT

Solved 200+ DSA Problems on LeetCode	Feb' 2026
MEDHA 2024 Hackathon: Achieved 5 th rank among 60 teams.	Aug' 2024

EDUCATION

Lovely Professional University	Phagwara, Punjab
Bachelor of Technology - Computer Science and Engineering – CGPA: 8.34	Aug' 2023 – Present
Touch Wood School	Dehradun, Uttarakhand
Intermediate – Percentage: 80	Apr' 2022 – Mar' 2023
Matriculation – Percentage: 90	Apr' 2020 – Mar' 2021